

Heart of Algebra



Solving linear equations

- 1 Solve for x : $4x - 9 = 23$.
 - 2 Solve for x : $\frac{3x + 2}{5} = 4$.
 - 3 Solve for x : $6(x - 3) = 2(x + 9)$.
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Linear inequalities

- 4 Solve $5x - 3 > 12$.
 - 5 Solve $-2x + 7 \leq 15$.
 - 6 Solve $3(x - 2) \geq x + 4$.
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Slope and equations of lines

- 7 Find the slope of the line through $(2, 5)$ and $(6, 13)$.
 - 8 Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.
 - 9 Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.
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Systems of equations by substitution

- 10 Solve the system: $y = 3x - 4$ and $2x + y = 11$.
 - 11 Solve the system: $x - 2y = 1$ and $3x + y = 17$.
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Systems of equations by elimination

- 12 Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.
- 13 Solve the system: $4x - y = 10$ and $2x + y = 14$, using elimination.

Absolute value equations

14 Solve $|2x - 5| = 9$.

15 Solve $|x + 4| = 10$.

Interpreting linear models

16 A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.

17 Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d -mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Systems with no solution or infinitely many solutions

18 For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

19 Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

Compound inequalities

20 Solve the compound inequality $-3 < 2x + 1 \leq 9$.

21 Solve $4 - 3x \geq 13$.

Multi-step word problems

22 Two numbers have a sum of 50 and a difference of 12. Find the two numbers.

23 The sum of three times a number and 7 equals 5 less than twice the number. Find the number.

A well-designed word problem: ticket sales at a community theater

- 24 This question has three parts. A community theater sells adult tickets for 15 dollars and child tickets for 8 dollars. On Saturday the theater sold 120 tickets total and collected 1,450 dollars. (a) Set up a system of two equations for the number of adult tickets a and child tickets c sold. (b) Solve the system to find how many of each ticket type were sold. (c) On Sunday the theater raises the adult price to 18 dollars but keeps the child price the same. If the same number of adult and child tickets sell as on Saturday, find the new total revenue.
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Multiple choice — easy

- 25 Solve for x : $-2x + 19 = 23$.

a. -4
b. 0
c. -2
d. 2

- 26 Solve for x : $-3x - 19 = 14$.

a. 11
b. -13
c. -9
d. -11

- 27 Solve for x : $2x + 11 = 13$.

a. -1
b. 3
c. 4
d. 1

28 Solve for x : $7x + 15 = 29$.

a. 0

b. 2

c. -2

d. 4

29 Solve for x : $4x - 14 = 2$.

a. 6

b. -4

c. 4

d. 2

30 Solve for x : $6x - 9 = -51$.

a. -7

b. -5

c. -9

d. 7

31 Solve for x : $6x - 2 = 16$.

a. 5

b. 3

c. -3

d. 1

32 Solve for x : $-9x + 20 = 119$.

a. -11

b. -13

c. -9

d. 11

33 Solve for x : $-3x - 19 = 26$.

a. -17

b. -15

c. 15

d. -13

34 Solve for x : $-8x + 1 = 9$.

a. 2

b. -3

c. 1

d. -1

35 Solve for x : $-6x + 3 = -33$.

a. 8

b. 4

c. 6

d. -6

36 Solve for x : $8x - 4 = -124$.

a. -17

b. 15

c. -15

d. -13

37 Solve for x : $-3x + 3 = -12$.

a. 7

b. 5

c. -5

d. 3

38 Solve for x : $4x - 11 = -15$.

a. 2

b. 1

c. -1

d. -3

39 Solve for x : $8x - 7 = 89$.

a. 14

b. 12

c. -12

d. 10

Multiple choice — medium

- 40 Solve the system: $4x + y = -14$ and $4x - y = 6$, using elimination.
- a. $(-1, -12)$
 - b. $(-1, -10)$
 - c. $(1, -10)$
 - d. $(-10, -1)$
- 41 Solve the system: $5x - 5y = -55$ and $5x + 5y = 5$, using elimination.
- a. $(-5, 6)$
 - b. $(6, -5)$
 - c. $(-3, 6)$
 - d. $(-5, 4)$
- 42 Solve the system: $2x + 5y = -1$ and $2x - 5y = -11$, using elimination.
- a. $(-3, 1)$
 - b. $(-3, -1)$
 - c. $(-1, 1)$
 - d. $(1, -3)$
- 43 Solve the system: $6x + 4y = -44$ and $6x - 4y = 20$, using elimination.
- a. $(-2, -10)$
 - b. $(-8, -2)$
 - c. $(-2, -8)$
 - d. $(0, -8)$

- 44 Solve the system: $3x - 3y = -21$ and $3x + 3y = 15$, using elimination.
- a. $(-1, 4)$
 - b. $(-1, 6)$
 - c. $(1, 6)$
 - d. $(6, -1)$
- 45 Solve the system: $2x + 4y = -6$ and $2x - 4y = 18$, using elimination.
- a. $(-3, 3)$
 - b. $(5, -3)$
 - c. $(3, -5)$
 - d. $(3, -3)$
- 46 Solve the system: $4x - 5y = 1$ and $4x + 5y = 71$, using elimination.
- a. $(9, 7)$
 - b. $(11, 7)$
 - c. $(7, 9)$
 - d. $(9, 5)$
- 47 Solve the system: $2x + 4y = 40$ and $2x - 4y = -24$, using elimination.
- a. $(8, 4)$
 - b. $(4, 6)$
 - c. $(4, 8)$
 - d. $(6, 8)$

- 48 Solve the system: $x - 2y = 6$ and $x + 2y = 6$, using elimination.
- a. $(6, -2)$
 - b. $(6, 0)$
 - c. $(8, 0)$
 - d. $(0, 6)$
- 49 Solve the system: $2x - 6y = 38$ and $2x + 6y = -70$, using elimination.
- a. $(-8, -9)$
 - b. $(-6, -9)$
 - c. $(-9, -8)$
 - d. $(-8, -11)$
- 50 Solve the system: $3x + y = -17$ and $3x - y = -37$, using elimination.
- a. $(10, -9)$
 - b. $(-7, 10)$
 - c. $(-9, 10)$
 - d. $(-9, 8)$
- 51 Solve the system: $x - y = 17$ and $x + y = -1$, using elimination.
- a. $(-9, 8)$
 - b. $(8, -11)$
 - c. $(8, -9)$
 - d. $(10, -9)$

52 Solve the system: $x - 5y = -48$ and $x + 5y = 52$, using elimination.

- a. (2, 8)
- b. (4, 10)
- c. (2, 10)
- d. (10, 2)

53 Solve the system: $6x - y = -56$ and $6x + y = -40$, using elimination.

- a. (-8, 6)
- b. (-8, 8)
- c. (8, -8)
- d. (-6, 8)

54 Solve the system: $5x - y = -24$ and $5x + y = -6$, using elimination.

- a. (-1, 9)
- b. (-3, 7)
- c. (-3, 9)
- d. (9, -3)

Multiple choice — hard

55 Solve $|2x - 9| = 25$.

- a. 8, -17
- b. 17
- c. -8, 18
- d. -8, 17

56 Solve $|2x - 12| = 26$.

- a. 19
- b. $-7, 19$
- c. $7, -19$
- d. $-7, 20$

57 Solve $|3x| = 18$.

- a. $6, -6$
- b. $-6, 6$
- c. $-6, 7$
- d. 6

58 Solve $|2x + 12| = 14$.

- a. 1
- b. $-13, 2$
- c. $-13, 1$
- d. $13, -1$

59 Solve $|2x - 7| = 21$.

- a. 14
- b. $7, -14$
- c. $-7, 14$
- d. $-7, 15$

60 Solve $-5x + 11 \geq 1$.

a. $x \leq 2$

b. $x \leq 0$

c. $x \geq 2$

d. $x \leq 4$

61 Solve $-4x - 2 \geq -42$.

a. $x \geq 10$

b. $x \leq 10$

c. $x \leq 12$

d. $x \leq 8$

62 Solve $-8x + 11 \geq 11$.

a. $x \leq -2$

b. $x \geq 0$

c. $x \leq 2$

d. $x \leq 0$

63 Solve $-7x - 3 \geq 53$.

a. $x \leq -8$

b. $x \leq -10$

c. $x \leq -6$

d. $x \geq -8$

64 Solve $-6x - 12 \geq 48$.

- a. $x \leq -8$
- b. $x \geq -10$
- c. $x \leq -12$
- d. $x \leq -10$

65 Two numbers have a sum of 63 and a difference of 9. Find the larger number.

- a. 38
- b. 36
- c. 34
- d. 27

66 Two numbers have a sum of 114 and a difference of 16. Find the larger number.

- a. 63
- b. 49
- c. 67
- d. 65

67 Two numbers have a sum of 31 and a difference of 7. Find the larger number.

- a. 17
- b. 12
- c. 19
- d. 21

68 Two numbers have a sum of 147 and a difference of 15. Find the larger number.

a. 81

b. 79

c. 66

d. 83

69 Two numbers have a sum of 146 and a difference of 4. Find the larger number.

a. 71

b. 75

c. 73

d. 77